

Demonstration of Proposed Features

Date	15 March 2026
Team ID	TM-RISINGWATERS-04
Project Name	Rising Waters: AI Powered Flood Prediction System
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Flood Prediction Module	Custom PyTorch UNet deep learning model processing Sentinel-2 multispectral bands and historical rain metrics to predict surface inundation overlays.	Implemented	Yes	Successfully achieved an intersection-over-union (IoU) of 86% on historical test basins.
2	Live Telemetry Dashboard	Dynamic FastAPI dashboard visualizing sensor readings (water depth, flow velocity, rainfall level) mapped with interactive Chart.js widgets.	Implemented	Yes	Connected live IoT node simulation threads showing stable continuous data streams.
3	Automated Twilio Dispatch	Twilio API pipeline triggering localized emergency SMS alerts directly to geo-fenced community contacts upon water level breach.	Implemented	Yes	Successfully demonstrated live message dispatch to registered phones during testing.
4	PostgreSQL / PostGIS Database	Robust backend storing geospatial layers, node records, user alerts, and handling highly optimized spatial intersection queries.	Implemented	Yes	Fully indexed GIS layers showing response times under 50ms for query overlaps.
5	Interactive GIS Inundation Map	Frontend visualizer displaying active vector shapes of prediction zones on Mapbox/Leaflet views with real-time opacity sliders.	Implemented	Yes	Rendered static and predictive water layers smoothly on browser interface.
6	Containerized Microservices	Configured Docker and docker-compose files for easy environmental consistency across database, FastAPI, and celery background worker.	Partial	Yes	Dev and QA environments containerized. Full scale Production container scaling is pending orchestration setup.

Feature Implementation Summary:

Total Features Proposed	6
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Total Features Implemented	5 (Fully Implemented) + 1 (Partially Implemented)
Total Features Demonstrated	6
Overall Implementation Rate (%)	91.67%